

		Angular speed, $\omega = \frac{v}{r} = 1.5 \times 10^4 \text{ rad s}^{-1}$
10	D	Since $v = r\omega$, $\omega = \frac{v}{r}$ and $a = \omega^2 r$ When r reduces to $\frac{1}{2}r$ and ω increases to 2ω, $v_{\text{new}} = 2\omega \left(\frac{1}{2}r\right) = v$ and $a_{\text{new}} = \left(\frac{1}{2}r\right)(2\omega)^2 = 2a$
11	B	Gravitational field strength due to mass M is directly proportional to the mass M and